

Handheld Ultrasonic Flowmeter

The **ATVUK SFM HH** ultrasonic flow meter is designed to measure the fluid velocity of liquid within a closed conduit. The transducers are a non-contacting, clamp-on type, which will provide benefits of non-fouling operation and easy installation.

The **ATVUK SFM HH** transit time flowmeter utilizes two transducers that function as both ultrasonic transmitters and receivers. The transducers are clamped on the outside of a closed pipe at a specific distance from each other.

The transducers can be mounted in V-method where the sound transverses the pipe twice, or W-method where the sound transverses the pipe four times, or in Z-method where the transducers are mounted on opposite sides of the pipe and the sound crosses the pipe once. This selection of the mounting method depends on pipe and liquid characteristics.

The flow meter operates by alternately transmitting and receiving a frequency modulated burst of sound energy between the two transducers and measuring the transit time that it takes for sound to travel between the two transducers. The difference in the transit time measured is directly and exactly related to the velocity of the liquid in the pipe



Feature :

High measuring Accuracy : Linearity: better than 0.5%,
Repeatability: better than 0.2%,
Accuracy: better than $\pm 1\%$

Small size, Light weight : Dimensions: 200×93×33mm; Weight: 390g

Wide measuring range : Several types transducer for selection,
Measuring pipe size from DN15mm to DN6000mm

Large LCD display : Display instantaneous flow, working status,
accumulated flow (positive, negative and net), velocity, etc.

Built-in data logger : Built-in 24K data logger,
store over 2000 lines measuring data

Non-intrusive measuring / Non contact measurement : Non-intrusive, clamp-on transducer,
no pressure drop, no pipe disturbance.

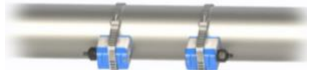
Power supply : When fully recharged it will last over 12 hours of operation,
3 AAA Ni-H built-in batteries, 100V-240VAC for the charger

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Optional Transducer Type :

Types	Picture	Spec.	Model	Measurement Range	Temperature	Dimension
Clamp on		Small Size	SS	DN15~DN100	-30~90℃	45×25×32mm
		Medium Size	MS	DN50~DN700	-30~90℃	64×39×44mm
		Large Size	LS	DN300~DN6000	-30~90℃	97×54×53mm
High Temperature Clamp on		Small Size	SS-HT	DN15~DN100	-30~160℃	45×25×32mm
		Medium Size	MS-HT	DN50~DN700	-30~160℃	64×39×44mm
		Large Size	LS-HT	DN300~DN6000	-30~160℃	97×54×53mm
Mounting Bracket Clamp on		Small Size	HSS	DN15~DN100	-30~90℃	318×59×85mm
		Medium Size	HMS	DN50~DN300	-30~90℃	568×59×85mm
		Extended	EX-1	DN300~DN700	-30~90℃	188×59×49mm
High Temperature Mounting Bracket Clamp on		Small Size	HSS-HT	DN15~DN100	-30~160℃	318×59×110mm
		Medium Size	HMS-HT	DN50~DN300	-30~160℃	568×59×110mm
		Extended	EX-1-HT	DN300~DN700	-30~160℃	188×59×49mm

Optional measuring types :

	<u>Clamp on transducer With mounting bracket</u> DN 15 to DN 700 Temp : -30 to 90/160
	<u>Clamp on transducer</u> DN 15 to DN 700 Temp : -30 to 90/160
	<u>Clamp on transducer</u> DN 15 to DN 6000 Temp : -30 to 160

Data sheet :

Convertor	Principle	Transit-time ultrasonic flowmeter
	Performance	Accuracy : ±1% Of reading at rates, Repeatability : better than 0.2% , Response time : 0 - 999 Second
	Display	4×16 character LCD with backlight, 7-digit totals for net, positive and negative flow respectively, display instant flow, velocity, total flow, signal status
	Communication	Insulate Rs232 serial interface,
	Data logger	Built-in-data logger, can store 2000 lines of data
Pipe Installation Condition	Material	Steel, Stainless steel, Cast iron, Copper, Cement pipe, PVC, Aluminum, Glass steel product, liner is allowed
	Size	25~6000mm
	Straight run	Transducer installation should be satisfied: upstream10D, downstream 5D, 30D from the pump
Measuring medium	Liquid type	Single liquid can transmit sound wave, such as Water (hot water, chilled water, city water, sea water, waste water, etc.); Sewage with small particle content; Oil (crude oil, lubricating oil, diesel oil, fuel oil, etc.); Chemicals (alcohol, etc.); Plant effluent; Beverage; Ultra-pure liquids, etc.
	Temperature	-30~ 90 /160℃
	Turbidity	No more than 10000ppm and less bubble
	Flow rate	0~±10m/s
Working Environment	Temperature	Convertor: -20~60℃ ; Flow Transducer: -30 ~ 90/160℃
	Humidity	Convertor: 85%RH; Flow Transducer: Ip67
Power Supply	Rechargeable battery	3 AAA built-in Ni-MH batteries 1.2V(for over 12 hours of operation). 90V-260V AC adapter
Power consumption		1.5W

Ordering code :

ATVUK-SFM-HH – Optional Transducer – Cable length

5 5m×2 (Standard)

10 10m×2

Example: ATVUK-SFM-HH - **SS-MS-LS** - **10m** (convertor with small, medium and large transducer + cable 5 meter + standard accessories)

PS: Standard configuration = Atvuk-SFM-HH (converter)+MS+5M*2 cable +Stretcher + Couplant(1 pc) + Carrying case + Ruler